

Responsible & Sustainable AI Report

Project: Smart City Traffic Analytics Capstone

1. Proxy Label Methodology

Because an explicit accident dataset was not provided, a proxy accident risk target (high_risk) was created by identifying records where **Severe Congestion** (traffic > 5,500 vehicles/hr) overlapped with **Adverse Weather Conditions** (snow, squalls, fog, or thunderstorm events).

2. Bias & Fairness Considerations

Extreme weather conditions are rare in historical records, creating severe class imbalance. Models trained on this data may exhibit higher false-negative rates during rare localized events. Additional environmental sensors across more geographic regions are recommended to reduce spatial sampling bias.

3. Environmental Sustainability & Efficiency

Lightweight tree-based ensembles (Random Forest / XGBoost) were selected over deep neural networks. They achieve low inference latency (< 10 ms) and require minimal computational power, allowing real-time deployment on edge-computing traffic controllers with a minimal carbon footprint.